One-To-One Functions

A function $f\colon A \to B$ associates a single $b \in B$ to each $a \in A$, but the same $b \in B$ may be associated to several values of A.

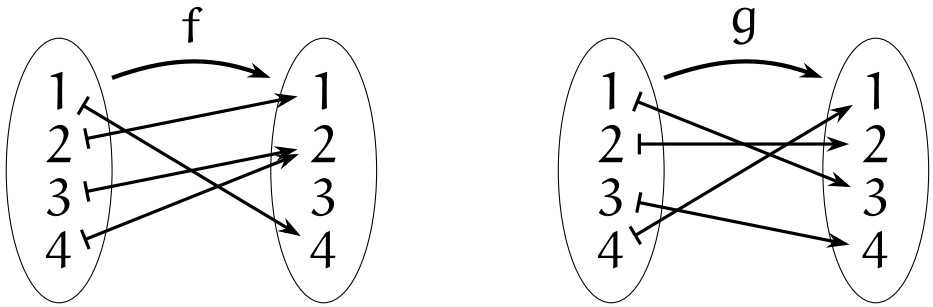
For example, for $f(x) = x^3 - x$ we have $f(-1)=0$ and $f(0)=0$ and $f(1) = 0$.

If f does *not* assign the same value to different inputs, it is called ***one-to-one*** (1-1), or ***injective***:

$$\forall\, a, a' \in A \left(a \neq a' \;\Rightarrow\; f(a) \neq f(a')\right).$$

`For all a ∈ A, a' ∈ A, if a ≠ a' then f(a) ≠ f(a').`

Example: Consider the following functions:



f is not injective, as $f(3) = f(4) = 2$.

However, g is injective.

One-To-One Functions

Exercise: Which of the following functions are one-to-one?

1. The function $\mathrm{score}\colon \mathrm{Class} \to \mathrm{Marks}$.

2. The function $f\colon \mathbb{N} \to \mathbb{N}$ defined by $f(x) = x^2$.

3. The function $f\colon \mathbb{R} \to \mathbb{R}$ defined by $f(x) = x^2$.

4. The function $f\colon \mathbb{R} \to \mathbb{R}$ defined by $f(x) = x^3$.

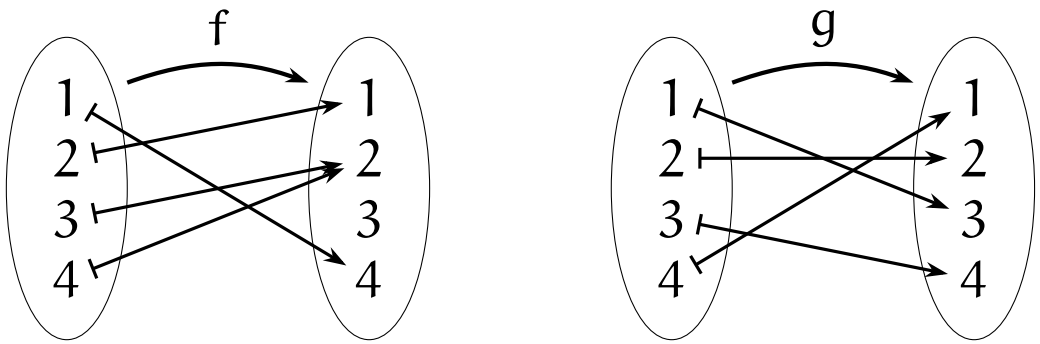
Onto Functions

A function $f: A \to B$ is ***onto***, or ***surjective***, if, and only if, every value in its codomain is produced by the function; that is, its range equals its codomain:

$$\forall\, b \in B\ \exists\, a \in A\ \Big(f(a) = b\Big).$$

`For all b ∈ B there exists a ∈ A such that f(a) = b.`

Example: Consider the following functions:



f is not surjective, as the value 3 is not produced by the function.

However, g is surjective.

Onto Functions

Exercise: Which of the following functions are onto?

1. The function $\mathrm{score}\colon \mathrm{Class} \to \mathrm{Marks}$.

2. The function $f\colon \mathbb{N} \to \mathbb{N}$ defined by $f(x) = x^2$.

3. The function $f\colon \mathbb{R} \to \mathbb{R}$ defined by $f(x) = x^2$.

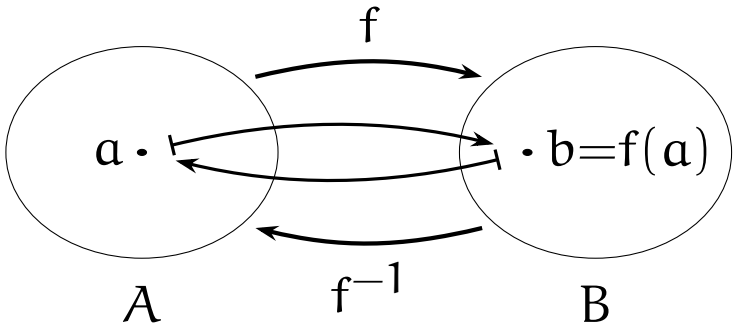
4. The function $f\colon \mathbb{R} \to \mathbb{R}$ defined by $f(x) = x^3$.

Bijective Functions

A function $f\colon A \to B$ is ***bijective***, or a ***bijection***, if, and only if, it is both one-to-one and onto.

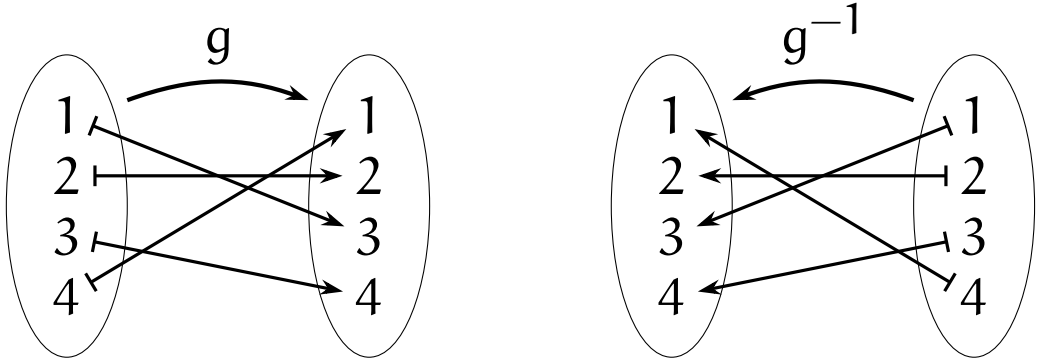
The arrows of a bijection can be reversed to give the ***inverse function*** $f^{-1}\colon B \to A$ that assigns to each $b \in B$ the unique $a \in A$ such that $f(a)=b$:

$$f^{-1}(b) = a \text{ if, and only if, } f(a)=b$$

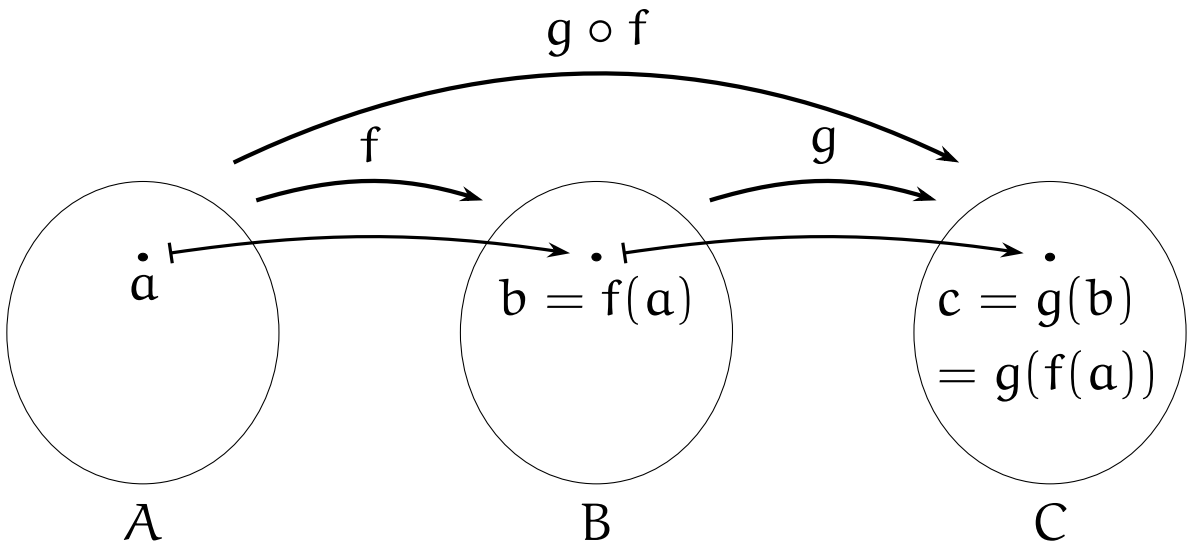


$f^{-1}\colon B \to A$ is also a bijection, and $(f^{-1})^{-1} = f$.

Example:



Properties of Composed Functions



- If $f : A \to B$ and $g : B \to C$ are both one-to-one,

  then so is $g \circ f : A \to C$.

- If $f : A \to B$ and $g : B \to C$ are both onto,

  then so is $g \circ f : A \to C$.

- If $f : A \to B$ and $g : B \to C$ are both bijective,

  then so is $g \circ f : A \to C$.